

Def: (geom. genus) C sm irr proj curve

$$g = h^0(C, \Omega_C/k) = h^1(C, \mathcal{O}_C)$$

e.g.

$$\textcircled{1} \Omega_{\mathbb{P}^1/\mathbb{C}} = \mathcal{O}(-2).$$

$$\mathbb{P}^1 = \text{Spec } k[x] \cup \text{Spec } k[x^{-1}].$$

$$dx^{-1} = x^{-2} dx.$$

$$\Rightarrow \Omega_{\mathbb{P}^1/x} = \mathcal{O}(-2).$$

$\textcircled{2}$ Hyperelliptic curves: $C \xrightarrow{\text{double cover}} \mathbb{P}^1$

Recall: $C' \xrightarrow{\text{double cover}} \mathbb{A}^1$, branch pts $p_1, \dots, p_r$,
 $\text{char } k \neq 2, k \neq \bar{k}$ $\text{Spec } k[x]$.

$$\Rightarrow C' = \text{Spec } k[x, y] / (y^2 - (x-p_1) \cdots (x-p_r)) = \text{Spec } k[x, y] / (y^2 - f(x))$$

$$\Omega_{C/k} = A dx \oplus A dy / (2y dy - f'(x) dx) \Rightarrow \text{invertible sheaf}$$

$$U = \frac{1}{x}, \quad P' = \underset{U_0}{\text{Spec } k[x]} \cup \underset{U_1}{\text{Spec } k[u]} \quad \left(\begin{array}{l} y \neq 0: \text{ generated} \\ \text{by } dx \\ y=0, \dots, dy \end{array} \right)$$

$$C_0 = \text{Spec } k[x, y] / y^2 - (x - p_1) \cdots (x - p_r)$$

$$C_1 = \text{Spec } k[u, v] / \underbrace{v^2 - (u - \frac{1}{p_1}) \cdots (u - \frac{1}{p_r})}_{"}$$

$$\frac{1}{x^r p_1 \cdots p_r} \left(p_1 \cdots p_r v^2 - x^r - (-1)^r (x - p_1) \cdots (x - p_r) \right)$$

$$\Rightarrow v^2 = \frac{(-1)^r y^2}{p_1 \cdots p_r x^r}$$

$\Rightarrow r$ is even.

Cech complex: $r = 2g + 2$

Degree of $\Omega_{C/k}$: $C = C_0 \cup C_1$

dx generates $\Omega_{C/k}$ on $(C_0)_y$

dy generates $\Omega_{C/k}$ on $(C_0)_{f(x)}$
 $y=0: X=p_i, i=1, \dots, r, 2ydy = f'(x)dx$

$\Rightarrow p_1, \dots, p_r$ are single zeros of dx .

On $C_1: dx = u^{-2} du$

$C_1 \setminus C_0 = \{u=0\} = \left\{ (u=0, v = \pm \sqrt{(1-1)^{\frac{r-1}{2}} \frac{1}{p_1 \dots p_r}}) \right\}$
 two double poles.

$\Rightarrow \deg \Omega_{C/k} = r - 4 = 2g - 2$

$H^0(C, \Omega_{C/k})$:

① $\forall w \in H^0(C_0, \Omega_{C/k}), w = a(x,y)dx + b(x,y)dy$.

Want: w is preserved by $y \mapsto -y$

$\Rightarrow w \in \text{Im}(\Omega_{\text{spectral}/k} \rightarrow \Omega_{C/k})$.

Pf: $y^2 = f(x) \Rightarrow dy = \frac{f'(x)}{2y} dx$.

$\Rightarrow w = \underbrace{\left(a(x,y) + b(x,y) \frac{f'(x)}{2y} \right)}_{\in k(x,y)} dx$.

$[k(x,y) : k(x)] = 2 \Rightarrow w = \frac{p(x)}{q(x)} dx, \gcd(p, q) = 1$.

dx generates $\Omega_{C/k}$ on $(C_0)_y \Rightarrow q(x) \neq 0$ if $x \notin \{p_1, \dots, p_r\}$

$p_1, \dots, p_r$ are single zeros of dx

$$\Rightarrow q(x) \mid f(x) = (x-p_1) \cdots (x-p_r)$$

$$x-p_1 = \frac{y^2}{(x-p_2) \cdots (x-p_r)} \Rightarrow p_1 \text{ is a pole of } \frac{dx}{x-p_1}$$

$$\Rightarrow q(x) = 1, \omega = p(x) dx$$

②. $\forall w \in H^0(C, \Omega_{C/k})$ preserved by $y \mapsto -y$, then

$w = 0$, for $w = p(x) dx$ on C_0 , $w = t(u) du$ on C_1

$$\Rightarrow \forall w \in H^0(C, \Omega_{C/k}). \quad i: C \rightarrow C, y \mapsto -y, x = x.$$

$$i^* w = w.$$

$$\textcircled{3} \quad \frac{dx}{y} : \text{on } C_0, y=0 \Rightarrow x = p_1, \dots, p_r$$

$$\frac{dx}{y} = \frac{2}{f'(x)} dy \Rightarrow \frac{dx}{y} \text{ regular on } p_1, \dots, p_r. \text{ no zeros. on } C_0$$

$$dx = u^{-2} du, \quad \frac{1}{y} = C \frac{u^{g+1}}{v}$$

$$\begin{aligned} v^2 &= (-1)^r \frac{y^2}{p_1 \cdots p_r x^r} \\ \frac{1}{y} &= \sqrt{(-1)^r \frac{1}{p_1 \cdots p_r x^r v^2}} \\ &= C \frac{1}{x^{g+1} v} = C \frac{u^{g+1}}{v} \end{aligned}$$

$$\Rightarrow \frac{dx}{y} = \frac{C \cdot u^{g-1} du}{v} \text{ on } C_1.$$

Likewise $\frac{du}{v}$ regular on C_1 , no zeros on C_1 .

The zeros of $\frac{dx}{y}$ on C_1 are $\{u=0\}$, order = $g-1$.

$$\Rightarrow \forall 0 \leq i < g, w_i = \frac{x^i dx}{y} \in H^0(C, \Omega_{C/k}).$$

④ $w_0, \dots, w_{g-1} \in H^0(C, \Omega_{C/k})$ linearly independent:

Let $\{p, q\} = \{x=0\} \subseteq C_0$.

$$\text{ord}(w_i)_p = i \quad \checkmark.$$

⑤ $H^0(C, \Omega_{C/k}) = \text{Span}_k \{w_0, \dots, w_{g-1}\}$:

$\forall w \in H^0(C, \Omega_{C/k}), \exists! a_0, \dots, a_{g-1} \text{ s.t.}$

$w = \sum_i a_i w_i$ vanishes at p , order $\geq g$.

By ②, $w = \sum_i a_i w_i$ vanishes at q , order $\geq g$.

$\Rightarrow w = \sum_i a_i w_i$ has more than $2g$ zeros

$$\deg \Omega_{C/k} = 2g-2 \Rightarrow w = \sum_i a_i w_i.$$

2/3-17.

$$(a) h^1(C, \Omega_{C/k}) = 1, H^1(C, \Omega_{C/k}) = \langle x^{-1} dx \rangle.$$

$$H^1(P^1, \Omega_{P^1/k}) \xrightarrow{\sim} H^1(C, \Omega_{C/k}).$$

$$(b) H^0(C, \Omega_{C/k}) \times H^1(C, \mathcal{O}_C) \longrightarrow H^1(C, \Omega_{C/k}).$$

$$\left\langle \frac{dx}{y}, x \frac{dx}{y}, \dots, x^{g+1} \frac{dx}{y} \right\rangle \times \left\langle \frac{y}{x}, \dots, \frac{y}{x^g} \right\rangle$$

$$\longrightarrow \langle x^{-1} dx \rangle.$$

Pf: (a) Riemann-Roch:

$$h^0 - h^1 = \deg - g + 1.$$

$$g - h^1 = 2g - 2 - g + 1 \Rightarrow h^1 = 1.$$

Cech complex:

$$0 \longrightarrow H^0(C_0, \Omega_{C/k}) \oplus H^0(C_1, \Omega_{C/k}) \longrightarrow H^0(C_0 \cap C_1, \Omega_{C/k}) \longrightarrow 0$$

$$\text{On } \{u=0\}, \text{ord}(dx) = -2, \text{ord}(dy) = -(g+2)$$

$$\text{ord}(x) = -1, \text{ord}(y) = -(g+1)$$

$$w = p_0(x) dx + \underline{q_0(x)} y dx, \quad t(x) | f(x)$$

$$t(x).$$

$$u = \frac{1}{x}, \quad v = C \cdot \frac{u^{g+1}}{y} = C \cdot \frac{1}{x^{g+1}y}$$

$$du = -x^{-2} dx, \quad dv = C \cdot \frac{-1}{x^{g+1}y^2} dy + C \cdot \frac{-(g+1)}{x^{g+2}y} dx.$$

$$P(u) du + \frac{q_1(u)}{t_1(u)} v du = C \cdot \frac{-f'(x)y dx}{2x^{g+1}f^2(x)} + C \cdot \frac{-(g+1)}{x^{g+2}y} dx.$$

$$= -P_1(x') x^{-2} dx - \frac{q_1(x')}{t_1(x')} \cdot \frac{C y}{x^{g+1}f(x)} \cdot \frac{1}{x^2} dx.$$

$$P_0(x) dx + \frac{q_0(x)}{t_0(x)} y dx + P_1(u) du + \frac{q_1(u)}{t_1(u)} v du = x' dx$$

$$\Rightarrow P_0(x) - P_1(x') x^{-2} = x' \quad \times$$

$$\Rightarrow H'(C, \Omega_{A/k}) = (x' dx). \quad \square$$

2.3.5. DVR (A, m, k) is a localization of a f.g k -alg R . t is a uniformizer of A .

Show $\Omega_{A/k} = A dt$.

pf: (a) $\Omega_{A/k}$ is a f.g A -mod:

$$R \text{ f.g. } k\text{-alg} \Rightarrow \Omega_{R/k} \text{ f.g. } k\text{-mod.}$$

$$\Omega_{A/k} = \Omega_{R/k} \otimes_k A \text{ f.g. } A\text{-mod.}$$

$$(b) \Omega_{A/k} = \langle dt \rangle:$$

$$\pi: A \rightarrow A/m = k, \forall a \in A, a - \pi(a) \in m. \\ \underbrace{a - \pi(a)} \in k \hookrightarrow A.$$

$$\Rightarrow a - \pi(a) = t \sigma(a) \Rightarrow a = \pi(a) + t \sigma(a)$$

$$\Rightarrow da = \sigma(a) dt + t d\sigma(a).$$

$$\Rightarrow \Omega_{A/k} \subseteq \langle dt \rangle + m \Omega_{A/k}$$

$$\Rightarrow \Omega_{A/k} = \langle dt \rangle.$$

$$(c) \Omega_{A/k} = A dt:$$

Suffices to show $\forall a \in A, a dt = 0 \Rightarrow a = 0$

$a = ut^m$, suffices to show $t^m dt \neq 0, \forall m \geq 1$.

$$A \twoheadrightarrow A/(t^m) \cong k[t]/t^m \Rightarrow \Omega_{A/k} \twoheadrightarrow \Omega_{(k[t]/t^m)/k}$$

$$\Omega_{(k[t]/t^m)/k} \neq 0 \Rightarrow \Omega_{A/k} \neq 0.$$

Assume $t^m dt = 0$, then $\dim_k \Omega_{A/k} < \infty$

$$\Omega_{A/k} = \Omega_{R/k} \otimes_k A \Rightarrow \Omega_{A/k} = 0. \text{ X.D.}$$

21.3.5. Proj space and the Euler exact seq:

A ring, $\exists$ exact seq: $0 \rightarrow \Omega_{\mathbb{P}^n_A/A} \rightarrow \mathcal{O}_{\mathbb{P}^n_A}(-1)^{\oplus (n+1)} \xrightarrow{\phi} \mathcal{O}_{\mathbb{P}^n_A} \rightarrow 0$

Dualizing: $0 \rightarrow \mathcal{O}_{\mathbb{P}^n_A} \xrightarrow{\phi^\vee} \mathcal{O}_{\mathbb{P}^n_A}(1)^{\oplus (n+1)} \rightarrow \mathcal{T}_{\mathbb{P}^n_A/A} \rightarrow 0$

pf:

Define $\phi: \mathcal{O}(-1)^{\oplus (n+1)} \rightarrow \mathcal{O}$

$$(s_0, \dots, s_n) \mapsto x_0 s_0 + \dots + x_n s_n$$

21.3.F: ϕ surj. Verify on $D(x_i)$.

$$\alpha: \Omega_{\mathbb{P}^n_A/A} \rightarrow \mathcal{O}_{\mathbb{P}^n_A}(-1)^{\oplus (n+1)}$$

$$f_1 d\left(\frac{x_1}{x_0}\right) + \dots + f_n d\left(\frac{x_n}{x_0}\right).$$

$$f_i d\left(\frac{x_i}{x_0}\right) \mapsto f_i \left(-\frac{x_i}{x_0^2} dx_0 + \frac{1}{x_0} dx_i \right). \quad \square.$$

21.3.H: $h^1(\mathbb{P}^2_k, \Omega_{\mathbb{P}^2_k/k}) > 0$, $\Omega_{\mathbb{P}^2_k/k}$ is not the direct sum of line bundles, can't be filtered into line bundles.

$$\text{pf: } 0 = H^0(\mathbb{P}^2, \mathcal{O}(-1)^{\oplus (n+1)}) \rightarrow H^0(\mathbb{P}^2, \mathcal{O}) = k$$

$$\rightarrow H^1(\mathbb{P}^2, \Omega_{\mathbb{P}^2/k}) \Rightarrow h^1(\mathbb{P}^2, \Omega_{\mathbb{P}^2/k}) > 0.$$

Assume $0 \rightarrow \mathcal{F} \rightarrow \Omega_{\mathbb{P}^2/k} \rightarrow \mathcal{L} \rightarrow 0$ where $\mathcal{L}$ is a line bundle. $h_2 - h_1 = \int \mathcal{L}_{\mathbb{P}^2/k}$

$\Rightarrow h^1(\mathbb{P}^2, \mathcal{F}) > 0$ since $h^1(\mathbb{P}^2, \mathcal{L}) = 0$ 12.

21.4. Riemann-Hurwitz formula:

Def: A finite morphism of integral schs $\pi: X \rightarrow Y$ is said to be separable if it is dominant, and $K(X)/K(Y)$ is separable

generically fin: $\forall$ generic pt $\hat{p}$ of irr components of Y , $\pi^{-1}(p)$ finite.

generically separable: generically fin + dominant + ext of func field separable (f^{sm})

21.4.2: $\pi: X \rightarrow Y$ generically separable, X, Y irr varieties

$\dim X = \dim Y = n$, Y sm, then

$$0 \rightarrow \pi^* \Omega_{Y/k} \xrightarrow{\phi} \Omega_{X/k} \rightarrow \Omega_{X/Y} \rightarrow 0$$

Pf: Y sm $\xrightarrow{(21.2.V)} \Omega_{Y/k}$ locally free, rank = n .

$\Rightarrow \pi^* \Omega_{Y/k}$ locally free, rank = n .

X irr variety $\Rightarrow$ Suffices to show ϕ_η is inj.
 (η is the generic pt of X)

$$\underbrace{\mathcal{O}_\eta \otimes \pi^* \Omega_{Y/k}}_{\dim=n} \xrightarrow{\phi_\eta} \underbrace{\mathcal{O}_\eta \otimes \Omega_{X/k}}_{\dim=n} \rightarrow \underbrace{\mathcal{O}_\eta \otimes \Omega_{X/Y}}_{\substack{\parallel \\ 0, \text{ for } K(X)/K(Y) \\ \text{separable}}} \rightarrow 0.$$

✓. □.

Not separable:

$$X = \text{Spec } \mathbb{F}_p[X], \quad Y = \text{Spec } \mathbb{F}_p[X^p]$$

$$\underbrace{\mathcal{O}_\eta \otimes \pi^* \Omega_{Y/\mathbb{F}_p}}_{\dim=1} \xrightarrow{\quad} \underbrace{\mathcal{O}_\eta \otimes \Omega_{X/\mathbb{F}_p}}_{\dim=1} \rightarrow \underbrace{\mathcal{O}_\eta \otimes \Omega_{X/Y}}_{\neq 0} \rightarrow 0$$

↑
not inj

21.4-A. X, Y sm varieties, $\dim=n$, $\pi: X \rightarrow Y$
 generically separable. Then the locus where π is
 not smooth is pure $\text{codim}=1$, and can be interpreted
 as an effective divisor.

p.f. In local coordinates we can define $\det \phi$.

$$(\Omega_{X/Y} \otimes \pi^* \Omega_{Y/k}) \rightarrow (\Omega_{X/k})$$

$$(X/Y)_\eta = 0 \text{ (separated)} \Rightarrow (Y)_\eta \neq 0$$

$$\Rightarrow \det \phi \neq 0.$$

$$\pi \text{ sm at } x \in X \Leftrightarrow (\det \phi)_x \neq 0 \quad (\text{Assume not empty})$$

$$X \text{ sm, } \forall x, \mathcal{O}_{X,x} \text{ UFD}$$

$$(\det \phi_x) \in \mathcal{O}_{X,x}, \text{ ht} = 1 \Rightarrow \text{prime codim} = 1$$

$$\det \phi \text{ regular} \Rightarrow \text{effective divisor.} \quad \square.$$

$$\text{Def: } \{x \in X \mid \pi \text{ not sm at } x\}: \text{ramification locus (divisor)}$$

$$\downarrow$$

$$\{y \in Y \mid y = \pi(x), \pi \text{ not sm at } x\}, \text{branch locus (divisor)}$$

$$\text{Rem: } \pi: \underbrace{X}_{\text{sm}} \rightarrow Y, \exists U \hookrightarrow X \text{ s.t. } \pi|_U: U \rightarrow \pi(U)$$

$$\text{isom, and } \text{codim}(X \setminus U) > 1, \text{ then } Y \text{ not sm}$$

$$\text{Now let } X, Y \text{ sm, } \dim X = \dim Y = 1, k = \bar{k}.$$

$\pi: X \rightarrow Y$, cl pt $p \in X$, $q = \pi(p)$.

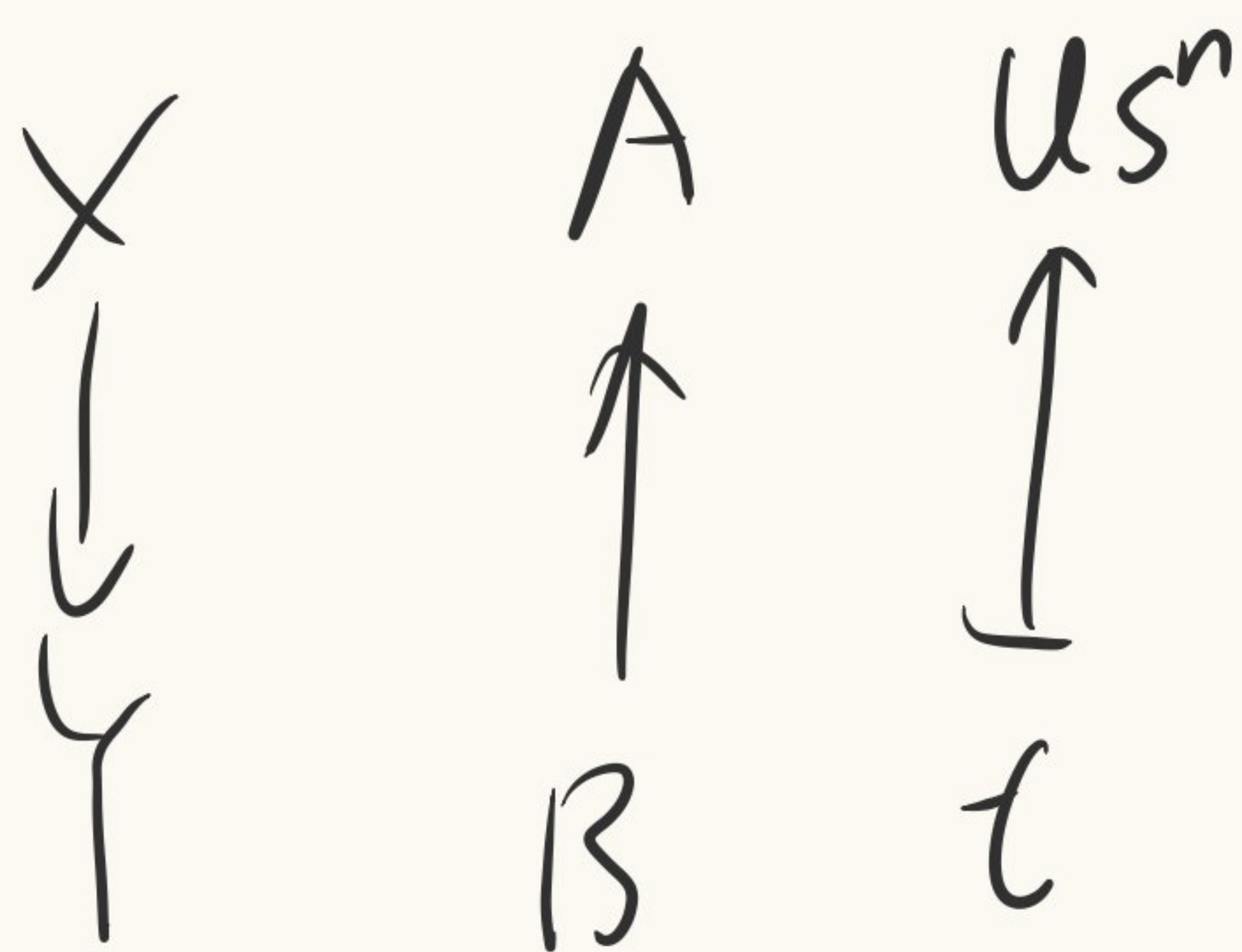
Wlog let $X = \text{Spec } \mathcal{O}_{X,p}$, $Y = \text{Spec } \mathcal{O}_{Y,q}$.

$\mathcal{O}_{X,p}, \mathcal{O}_{Y,q}$ DVR by X, Y sm.

$\begin{matrix} \mathcal{O}_{X,p} \\ \parallel \\ A \\ s \end{matrix}$

$\begin{matrix} \mathcal{O}_{Y,q} \\ \parallel \\ B \\ t \end{matrix}$

uniformizer:



$$\Rightarrow dt = d(us^n) = uns^{n-1}ds + s^n du.$$

vanish order: $n-1$: tame ramification

$\geq n$: wild ramification
(char $k | n$)

Def: (ramification index at p) = vanish order + 1.

21.4.B. Degree of $\Omega_{X/Y}$ at $p \in X$ = (ramification order of π at p) - 1

pf: Separable $\Rightarrow (\Omega_{X/Y})_n = \Omega_{K(X)/K(Y)} = 0$.

$\Rightarrow \Omega_{X/Y}$ is a torsion sheaf

deg of $\Omega_{X/Y}$ at p = length of $(\Omega_{X/Y})_p$.
 $= \dim_k (\Omega_{X/Y})_p$

$$0 \rightarrow \pi^* \Omega_{Y/k} \rightarrow \Omega_{X/k} \rightarrow \Omega_{X/Y} \rightarrow 0.$$

$dt \mapsto u ds^{n-1} + s^n du.$

$\Rightarrow \dim_k (\Omega_{X/Y})_p = \dots \dots \dots \quad \square$

Thm (Riemann-Hurwitz formula)

$\pi: X \rightarrow Y$, fin. separable, X, Y proj regular curves,
 π of pure degree n . R is the ramification divisor.

Then $2g(X) - 2 = n(2g(Y) - 2) + \deg R$.

pf: $0 \rightarrow \pi^* \Omega_{Y/k} \rightarrow \Omega_{X/k} \rightarrow \Omega_{X/Y} \rightarrow 0$.

$$\deg \Omega_{X/k} = \underbrace{\deg \pi^* \Omega_{Y/k}}_{n \deg \Omega_{Y/k}} + \deg \Omega_{X/Y}$$

Cor: $\pi: X \rightarrow Y$, X, Y sm, irr, proj curves.
 $g(X) = 2$, $g(Y) = 3 \Rightarrow \pi$ is constant.

Def: étale: sm + relative $\dim = n$.

We say X is simply connected in the étale top if the only fin étale cover of a connected sch X is an isom.

e.g. $k = \bar{k}$, $\mathbb{P}_k^1$ is ---.

$\pi: X \rightarrow \mathbb{P}_k^1$ étale.

$$R-W: 2g(X) - 2 = (\deg \pi) \cdot (-2)$$

$$\Rightarrow \deg \pi = 1, g(X) = 0,$$

e.g. $k = \bar{k}$, $\text{char } k = 0$, A_k^1 is ---.

$$\pi: \underset{\text{Spec } A}{X} \rightarrow A_k^1 = \text{Spec } k[x].$$

$$\leadsto \bar{\pi}: \bar{X} \rightarrow \mathbb{P}_k^1$$

$$\text{char } k = 0 \Rightarrow \text{R-W: } 2g(\bar{X}) - 2 = n(2g(\mathbb{P}_k^1) - 2) + \deg R.$$

$$= -2n + \deg R$$

$$= -2n + n - |\pi^{-1}(\infty)|.$$

$$= -n - \underbrace{|\pi^{-1}(\infty)|}_{\geq 1} \geq -2.$$

$$\Rightarrow n = |\pi^{-1}(\infty)| = 1$$

$\Rightarrow \bar{\pi}$ is an isom. $\checkmark$.

e.g. Lüroth Thm:

$$g(X) = 0 \Rightarrow g(Y) = 0 \Rightarrow Y = \mathbb{P}^1$$

$$X \rightarrow Y \leadsto K(Y) \hookrightarrow K(X) = k(X)$$

i.e. $\forall$ subfield $K(Y)$ of $k(X)$ containing k is of the form $k(f(X))$.

Rem: Riemann surface: $\pi: C \rightarrow \mathbb{P}_\mathbb{C}^1$

$\leadsto$ R-W formula $\Rightarrow$ geom genus = top genus.

18.3.2
= Arith genus

21.4.9: C irr sm proj curve, $g(C) \geq 2$,

fin $G \subseteq \text{Aut } C \Rightarrow |G| \leq 84(g(C)-1)$.

Sketch of pf: $K(C)^G \hookrightarrow K(C)$

$\rightsquigarrow \pi: C \longrightarrow C', \deg \pi = |G|$.

$$R-W: 2g(C)-2 = |G| \left(2g(C')-2 + \sum_{i=1}^n \frac{r_i-1}{r_i} \right)$$

n branch pts, $p_1, \dots, p_n$,

p_i has ramification r_i .

21.5. Def: $\Omega_{X/Y}^i := \Lambda^i \Omega_{X/Y}$ sheaf of relative i -forms

21.5.A: X, X' birational irr sm proj

k -varieties, then $H^0(X, \Omega_{X/k}^i) \cong H^0(X', \Omega_{X'/k}^i)$

pf: $\phi: X \dashrightarrow X'$, $\phi|_U: U \rightarrow \phi(U)$ isom.

or $1 \leq R$: $\text{codim}(X \setminus U) \geq 2$.

$$\hookrightarrow \phi^*: H^p(X', \Omega_{X'/k}^i) \rightarrow H^p(U, \Omega_{X/k}^i)$$

Alg Hartog's lemma (B.J.19): $A = \bigcap_{p \text{ codim}=1} A_p$

(A integrally closed
Noe integral domain)

$$\hookrightarrow H^p(U, \Omega_{X/k}^i) \xrightarrow{\sim} H^p(X, \Omega_{X/k}^i)$$

$$\phi^*: H^p(X', \Omega_{X'/k}^i) \rightarrow H^p(X, \Omega_{X/k}^i)$$

$$\psi^*: H^p(X, \Omega_{X/k}^i) \rightarrow H^p(X', \Omega_{X'/k}^i)$$

$$\phi^*|_U, \psi^*|_U: \checkmark$$

□

(Birational invariant: $h^p(X, \Omega_{X/k}^i)$)

Def: X dim=n, sm k -variety.

canonical bundle: $K_{X/k} := \det \Omega_{X/k} = \Omega_{X/k}^n$

Rem: If X proj, $K_{X/k} = \omega_X$.

21.5.13: X sm variety, Z sm sub variety

Then $K_Z \cong K_X|_Z \otimes \det \mathcal{N}_{Z/X}$. (Adjunction formula)

pf: 2.2.31: $i: Z \hookrightarrow X$

$$0 \rightarrow \mathcal{N}_{Z/X}^\vee \rightarrow i^* \Omega_{X/k} \rightarrow \Omega_{Z/k} \rightarrow 0.$$

14.2.p: $0 \rightarrow \mathcal{F}_1 \rightarrow \dots \rightarrow \mathcal{F}_n \rightarrow 0$, fin rank, locally free sheaves on X .

$$\Rightarrow \det \mathcal{F}_1 \otimes \det \mathcal{F}_2^\vee \otimes \dots \otimes \det(\mathcal{F}_n)^{(-1)^n} \cong \mathcal{O}_X$$

$$\det(\Omega_{Z/k})^\vee \otimes \det(\Omega_{X/k})|_Z \otimes \det \mathcal{N}_{Z/X} \cong \mathcal{O}_X. \quad \square$$

Def: X proj sm k -variety

$p_g(X) = h^0(X, K_X) = h^0(X, \Omega_{X/k}^n)$ is called geom genus.

e.g: Z is a regular $\deg = d$ surface in $\mathbb{P}_k^3$, $k = \bar{k}$.

Compute $p_g(Z)$:

$$h^0(Z, K_Z) = h^0(Z, (K_{\mathbb{P}^3}|_Z) \otimes \det \mathcal{N}_{Z/\mathbb{P}^3})$$

$$\mathcal{I}_Z = \mathcal{O}_{\mathbb{P}^3}(-Z) = \mathcal{O}(-d)$$

$$\mathcal{N}_{Z/\mathbb{P}^3}^\vee = \mathcal{I}_Z / \mathcal{I}_Z^2|_Z = \mathcal{I}_Z \otimes_{\mathcal{O}_X} \mathcal{O}_Z = \mathcal{O}_Z(-d).$$

$$\Rightarrow \mathcal{N}_{Z/\mathbb{P}^3} = \mathcal{O}(d)|_Z.$$

$$0 \rightarrow \mathcal{S}_{\mathbb{P}^3}^1 \rightarrow \mathcal{O}_{\mathbb{P}^3}(-1)^{\oplus 4} \rightarrow \mathcal{O}_{\mathbb{P}^3} \rightarrow 0.$$

$$\Rightarrow K_{\mathbb{P}^3} \otimes \mathcal{O}_{\mathbb{P}^3} = \mathcal{O}(-4)$$

$$K_{\mathbb{P}^3} = \mathcal{O}(-4).$$

$$\Rightarrow h^0(Z, \mathcal{K}_Z) = h^0(Z, \mathcal{O}(d-4)|_Z).$$

$$0 \rightarrow \mathcal{I}_Z \xrightarrow{\mathcal{O}(-d)} \mathcal{O}_{\mathbb{P}^3} \rightarrow \mathcal{O}_Z \rightarrow 0.$$

$$\Rightarrow 0 \rightarrow \mathcal{O}(-4) \rightarrow \mathcal{O}(d-4) \rightarrow \mathcal{O}_Z(d-4) \rightarrow 0$$

$\parallel$
 $\mathcal{K}_Z$

$$\Rightarrow 0 \rightarrow H^0(\mathbb{P}^3, \mathcal{O}(d-4)) \rightarrow H^0(Z, \mathcal{K}_Z) \rightarrow 0.$$

$$\Rightarrow P_g(Z) = \begin{cases} \binom{d-1}{d-4} & d \geq 4 \\ 0 & d \leq 3. \end{cases}$$

✓ Quartic surface Z in $\mathbb{P}^3$, $P_g(Z) = \binom{4-1}{4-4} = 1$

$$K_{\mathbb{P}^2} = \mathcal{O}(-3), \quad P_g(\mathbb{P}^2) = h^0(\mathbb{P}^2, \mathcal{O}(-3)) = 0$$

$\Rightarrow Z$ is not rational.

□

Def: X sm proj k -variety, X is Fano if $K_X^\vee$ is ample, Calabi-Yau if $K_X \cong \mathcal{O}_X$.

